FASHION FORWARD PROJECT REPORT

Customer Purchasing Behaviour and Product Demand Analysis

Project Category: Data Analytics

Tool Used: Microsoft Excel 2019

Project Type: Sales and Customer Behaviour Analysis

1. Project Overview

Fashion Forward is a Nigerian online fashion retail business used as the case study for this project. The business sells fashion products across four major categories: Dresses, Footwear, Bags, and Accessories. It receives orders through three sales channels: Website, Instagram, and WhatsApp.

The purpose of this project was to analyze the business sales transactions to understand its revenue performance, product demand, customer purchasing behaviour, geographical performance, monthly sales trends, and sales-channel performance.

Using Microsoft Excel 2019, I cleaned and prepared the dataset, performed exploratory and descriptive analysis using PivotTables, created visualizations with PivotCharts, and developed an interactive dashboard to communicate the findings.

The analysis was guided by seven business questions designed to provide useful insights that could support business decisions.

2. Project Objectives

The objectives of this project were to:

- Evaluate the overall sales performance of Fashion Forward.
- Identify the products with the highest sales quantities.
- Determine the revenue contribution of each product category.
- Compare the purchasing behaviour of new and returning customers.
- Evaluate sales performance across selected Nigerian states.
- Examine monthly revenue, order volume, and product demand.
- Compare the performance of the Website, Instagram, and WhatsApp sales channels.

3. Dataset Description

The dataset contained transaction-level information relating to Fashion Forward's sales activities.

The available fields included:

Order ID, Order Date, Customer Name, Customer Segment, Product, Product Category, Quantity, Unit Price, Revenue, State, Order Type, Payment Method.

The dataset contained inconsistencies, missing values, duplicate records, spelling variations, and differences in data formatting. These issues required attention before analysis.

After data cleaning, 660 transaction records were retained for the analysis.

4. Data Cleaning and Preparation

Data cleaning was carried out in Microsoft Excel 2019 to improve data consistency and ensure that the dataset was suitable for analysis.

The following activities were performed:

- Duplicate Records: Nine exact duplicate transaction rows were identified and removed to prevent repeated records from affecting the analysis.
- Missing Revenue Values: Eighteen missing Revenue values were identified. These were calculated using:
- Revenue = Quantity × Unit Price.
- Existing revenue values were retained.
- Missing Customer Names: Eight missing customer names were identified. These were left blank rather than replaced with invented information.
- Missing Payment Methods: Seven missing payment method entries were assigned the label "Unknown" because the original values could not be established.
- Inconsistent State Names: State entries were standardized into Lagos, Abuja, Rivers, and Oyo. Spelling variations such as "Abujaa" and "Riverz" were corrected.
- Product Categories: Inconsistent category names, including "Dresss", "Footware", "Bagss", and "Accesories", were standardized to Dresses, Footwear, Bags, and Accessories.
- Customer Segments: The spelling variation "Returnning Customer" was corrected to "Returning Customer".
- Order Types: Variations such as "Instagam" and "Whatsap" were corrected to Instagram and WhatsApp.
- Unit Price Formatting: Unit prices containing commas and stored as text were converted into numeric values to support calculations and analysis.

After cleaning, the dataset was checked to ensure that the relevant categories were standardized and that Revenue and Unit Price were in appropriate numeric formats.

5. Data Analysis and Findings

PivotTables were used to summarize the cleaned dataset and answer the seven business questions. PivotCharts were used to present selected findings visually.

Question 1: What is the overall sales performance?

The following key performance indicators were calculated:

Metric	Result
Total Revenue	₦29,156,500
Total Orders	660
Total Quantity Sold	2,036
Average Order Value	₦44,176.52

Average Order Value (AOV) was calculated by dividing total revenue by the total number of orders.

AOV = Total Revenue ÷ Total Orders.

The business generated ₦29,156,500 from 660 transactions, with an average order value of ₦44,176.52.

Question 2: Which five products recorded the highest quantities sold?

The top five products by quantity sold were:

Product	Quantity Sold
Bodycon Dress	215
Wrap Dress	169
Sneakers	160
Loafers	159
Ankara Gown	156

Bodycon Dress recorded the highest quantity sold, followed by Wrap Dress and Sneakers.

These products represent the highest-volume products in the dataset and may be useful when reviewing inventory and product availability.

Question 3: What percentage of total revenue came from each product category?

The revenue contribution of the four product categories was:

Product Category	Revenue	Revenue Share
Dresses	₦10,996,500	37.72%
Footwear	₦8,508,000	29.18%

Bags	₦6,537,500	22.42%
Accessories	₦3,114,500	10.68%

The above results indicates that Dresses and Footwear were the leading revenue-generating categories during the period analyzed.

Question 4: How do new and returning customers compare?

The analysis compared revenue, number of orders, and average order value across the two customer segments.

Customer Segment	**Revenue**	**Orders**	**Average Order Value**
New Customers	₦17,080,000	390	₦43,794.87
Returning Customers	₦12,076,500	270	₦44,727.78

New customers generated more revenue and accounted for more orders than returning customers.

However, returning customers recorded a slightly higher average order value of ₦44,727.78 compared with ₦43,794.87 for new customers. Returning customers accounted for approximately 41% of all orders. This suggests that repeat purchasing contributes meaningfully to sales, although new customers accounted for the larger share of transactions.

Question 5: How does sales performance compare across the four states?

The geographical analysis produced the following results:

State	Revenue	Orders
Lagos	₦8,386,000	185
Abuja	₦7,873,500	184
Rivers	₦6,756,000	154
Oyo	₦6,141,000	137

Lagos recorded the highest revenue and number of orders, while Oyo recorded the lowest figures.

An interesting observation is that Lagos and Abuja had almost the same number of orders, but Lagos generated higher revenue. This shows that order volume alone does not determine total revenue.

Question 6: What is the monthly revenue, order, and quantity trends?

The monthly analysis examined revenue, number of orders, and quantity sold from January to December.

Key findings included:

- November recorded the highest monthly revenue of ₦2,805,000.
- April recorded the lowest monthly revenue of ₦1,708,500.
- November recorded the highest number of orders at 67.
- April recorded the lowest number of orders at 43.
- July recorded the highest quantity sold at 197 units.
- April recorded the lowest quantity sold at 125 units.

The results show variations in monthly sales performance and product demand.

November was the strongest month in terms of revenue and order volume, while April recorded the lowest performance across the three measures.

Question 7: How do the three sales channels compare?

The analysis compared Website, Instagram, and WhatsApp using the number of orders and average order value.

Sales Channel	Number of Orders	Average Order Value
Website	270	₦43,490.74
Instagram	201	₦45,442.79
WhatsApp	189	₦43,809.52

The Website recorded the highest number of orders, making it the most-used sales channel by transaction volume. Instagram recorded the highest average order value at ₦45,442.79.

This indicates that although the Website generated the highest order volume, Instagram customers spent slightly more per transaction on average.

6. Key Business Recommendations

Based on the findings, the following recommendations are proposed:

i. Prioritize High-Demand Products and Categories

Fashion Forward should pay attention to the availability of products such as Bodycon Dress, Wrap Dress, Sneakers, Loafers, and Ankara Gown. Dresses and Footwear should also receive attention during inventory planning because they contributed the largest shares of revenue.

ii. Strengthen Customer Retention

Since returning customers recorded a slightly higher average order value, the business could explore customer retention strategies such as loyalty rewards, personalized product recommendations, and follow-up communication to encourage repeat purchases.

iii. Improve Sales and Inventory Planning

The business could use its geographical and monthly sales patterns to guide promotional activities and inventory planning. Lagos recorded the highest revenue, while November was the strongest month by revenue and order volume. These patterns can inform future planning, although further analysis would be needed to establish the causes of the differences.

7. Dashboard Development

An interactive Excel dashboard was developed to present the major findings in one place.

The dashboard contains four KPI cards: Total Revenue, Total Orders, Total Quantity Sold, Average Order Value

It also contains visualizations covering: Top five products by quantity sold, Revenue share by product category, Revenue by customer segment, Revenue by state, Monthly revenue trend, Order type distribution.

Four slicers were included to allow users to filter the dashboard by: State, Product Category, Customer Segment, Order Type

The dashboard was designed to make the findings easier to interpret and allow users to explore different aspects of the business performance.

8. Tools and Skills Applied

Software: Microsoft Excel 2019.

Excel Features Used: Data cleaning and preparation, Duplicate removal, Data standardization, Basic formulas and calculations, PivotTables, PivotCharts, Dashboard development, Slicers and interactive filtering.

Skills Demonstrated:

- Data preparation and organization.
- Exploratory and descriptive data analysis.
- Business question interpretation.
- Data summarization.
- Data visualization.
- Insight generation.
- Business recommendations.

- Communicating analytical findings.

9. Limitations

The analysis was based on the information available in the supplied dataset.

Missing customer names were not reconstructed, and missing payment methods were classified as Unknown.

Missing revenue values were calculated using Quantity multiplied by Unit Price, based on the consistency observed in the existing revenue records.

Average Order Value was calculated using total revenue divided by the number of transaction records. This calculation assumes that each cleaned transaction row represents one order.

The analysis focused on sales revenue and purchasing behaviour. It did not include expenses, cost of goods sold, profit margins, advertising expenditure, or customer acquisition costs. Therefore, the findings should not be interpreted as a complete measure of business profitability or marketing effectiveness.

10. Conclusion

The Fashion Forward project provided an opportunity to apply Microsoft Excel to a practical business analysis scenario.

Through data cleaning, PivotTables, PivotCharts, and dashboard development, I was able to examine sales performance, identify high-demand products, understand customer purchasing patterns, compare geographical performance, and evaluate sales channels.

The findings showed that Dresses contributed the largest share of revenue, Bodycon Dress recorded the highest quantity sold, Lagos generated the highest revenue, November was the strongest month by revenue, and the Website recorded the highest number of orders. Instagram, however, recorded the highest average order value.

Overall, this project strengthened my understanding of how data can be transformed into meaningful insights that support business analysis and decision-making.